Cyber Threat Monitoring Level I (CTM-L1)

Institutional Assessment

"Network Intrusion Detection and Initial Incident Response Assessment"

Name of Trainee: Aldrietch Dakay

Name of Training Institute: A1 Ventures Technical and Vocational Academy Inc. - Carmen Cebu Branch

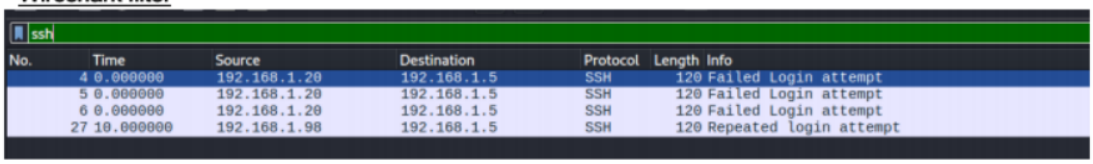
Part I. Detailed Analysis, evidences, interpretations of:

(Add Screenshots of evidences and provide detailed explanations for each interpretations.) You can modify the template according to your report.

FINDINGS 1: BRUTE FORCE ATTACK

Threat	Packet(s)	Analysis
Failed logins → possible brute force	Packets 4 – 6 and 27	Three failed SSH login attempts from 192.168.1.20 to 192.168.1.5, followed later by a repeated login attempt from 192.168.1.98. The pattern is consistent with authentication abuse/ reconnaissance. Correlate with SSH/authentication logs to determine whether any login succeeded.

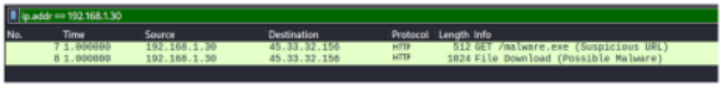
Wireshark filter



FINDINGS 2: MALWARE

Threat	Packet(s)	Analysis
Suspicious URLs → malware download	Packets 7 – 8	Packet 7 records GET /malware.exe and packet 8 records a possible malware download. This is a strong indicator of a potentially malicious file download.

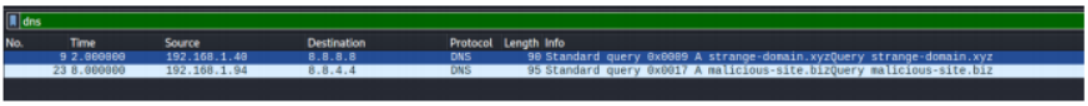
Wireshark filter screenshot:



FINDINGS 3: SUSPICIOUS DNS

Threat	Packet(s)	Analysis
Suspicious DNS	Packets 9 and 23	Queries for strange-domain.xyz and malicious-site.biz. Domains are suspicious

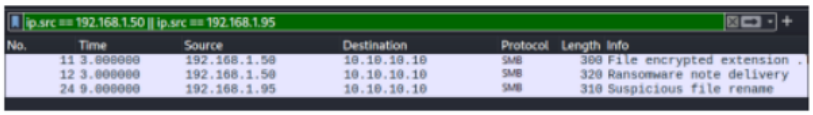
Wireshark filter screenshot:



FINDINGS 4: RANSOMEWARE

Threat	Packet(s)	Analysis
Ransomware	Packets 11, 12 and 24	Packet 11 records an encrypted .locked file, packet 12 a ransomware note delivery, and packet 24 a suspicious file rename. Together these are strong exercise indicators of file-encryption activity.

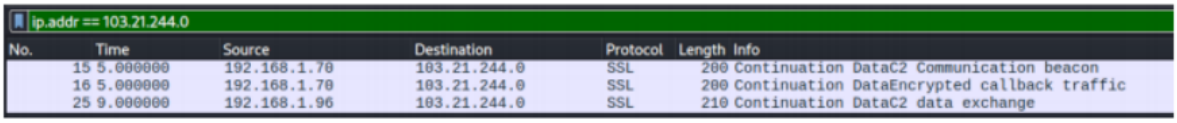
Wireshark filter screenshot:



FINDINGS 5: C2 COMMUNICATIONS

Threat	Packet(s)	Analysis
C2 communications	Packets 15 - 16 and 25	Packet 15 is a C2 beacon, packet 16 an encrypted callback, and packet 25 C2 data exchange. Repeated outbound communication is consistent with possible command-and-control behavior; validate destination, frequency, and endpoint processes.

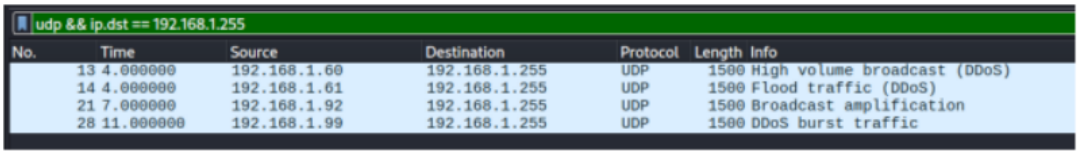
Wireshark filter screenshot:



FINDINGS 6: DDOS

Threat	Packet(s)	Analysis
DDoS activity → flood traffic	Packets 13, 14, 21 and 28	High-volume broadcast, flood traffic, broadcast amplification and a DDoS burst are recorded. This indicates potential flood/amplification behavior . Confirm actual service impact and whether the traffic is internal, reflected, or externally sourced.

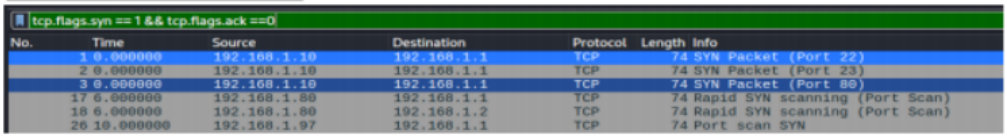
Wireshark filter screenshot:



FINDINGS 7: PORT SCANNING

Threat	Packet(s)	Analysis
Port scanning → reconnaissance	Packets 1 - 3, 17 - 18 and 26	SYN activity targets ports 22, 23 and 80, while later packets are labeled rapid SYN scanning and port-scan SYN. This is consistent with service discovery/reconnaissance. Identify the scanning host and scan scope.

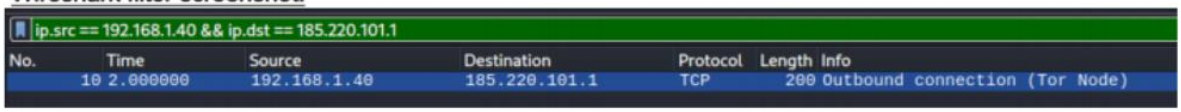
Wireshark filter screenshot:



FINDINGS 8: DATA EXFILTRATION

Threat	Packet(s)	Analysis
Possible data exfiltration / suspicious IPs	Packets 10	Packet 10 is notable because it identifies the destination as a Tor node. Tor traffic is not inherently malicious, but an unexpected outbound Tor connection from an internal workstation should be investigated

Wireshark filter screenshot:



SOC INCIDENT TICKET

Ticket Information

Ticket ID: SOC-CTM-L1-2026-02

Date Created: 10/02/2026

Reported By: TRACENYX

Assigned To: Level II Incident Handler

Incident Status: Escalated

Incident Summary

Multiple indicators of compromise were identified in the supplied network packet data. Evidence includes repeated SSH failed login attempts, TCP SYN port scanning, suspicious malware download activity, suspicious outbound connections, C2 beaconing and encrypted callback traffic, ransomware indicators including encrypted files and ransom note delivery, possible data exfiltration, and DDoS/flood traffic. Packet-level evidence was documented for escalation to Level II Incident Handling.

Severity Level: CRITICAL

Severity Justification: CRITICAL due to active file-encryption/ransomware indicators (packets 11 – 12 and 24) together with external communication/C2 indicators (packets 15 – 16 and 25). In a production environment, confirmation of these behaviors would warrant immediate containment and Level II handling because of potential confidentiality, integrity, and availability impact.

CATEGORY	DESCRIPTION
Attack Type	Multi-stage network intrusion involving reconnaissance/port scanning, possible SSH brute-force activity, malware download, suspicious outbound communication, ransomware activity, C2 communication, possible data exfiltration, and DDoS/flood traffic.
Affected Systems	Multiple internal hosts including 192.168.1.10, 192.168.1.20, 192.168.1.30, 192.168.1.40, 192.168.1.50, 192.168.1.60, 192.168.1.61, 192.168.1.70, 192.168.1.80, 192.168.1.90, 192.168.1.91, 192.168.1.92, 192.168.1.93, 192.168.1.94, 192.168.1.95, 192.168.1.96, 192.168.1.97, 192.168.1.98, and 192.168.1.99.
External IPs Involved	45.33.32.156 — malware download/callback; 185.220.101.1 — outbound connection identified in the data as a Tor node; 103.21.244.0 — C2 beacon/callback/data exchange; DNS queries were also sent to 8.8.8.8 but the supplied data does not identify these DNS servers themselves as malicious.
Detection Method	SIEM alerts followed by packet-level analysis in Wireshark. The assessment states that unusual traffic generated SIEM alerts and that the provided PCAP was analyzed to identify threats, correlate indicators, and document packet-level evidence.
Time Detected	May 29, 2026, 09:30:00 – 09:30:11. The first suspicious activity appears at 09:30:00, with the final listed event at 09:30:11.

INITIAL FINDINGS (LEVEL I ANALYSIS)

Port Scanning: Multiple TCP SYN scans detected from 192.168.1.10, 192.168.1.80 and 192.168.1.97 targeting internal hosts, indicating reconnaissance activity. Evidence: Packets 1 – 3, 17 – 18 and 26.

Brute Force Attempt: Multiple failed SSH login attempts detected from 192.168.1.20 to 192.168.1.5, indicating a possible SSH brute-force attack. Evidence: Packets 4 – 6.

Malware Activity: Host 192.168.1.30 requested and downloaded /malware.exe from 45.33.32.156. Evidence: Packets 7 – 8 .

Outbound Traffic: Suspicious outbound connection and HTTP POST data upload to external IPs. Packets 10, 19

Ransomware Indicators: .locked file encryption, ransomware note and suspicious file rename. Packets 11 – 12, 24.

DDoS Activity: UDP broadcast/flood, amplification and burst traffic toward 192.168.1.255. Packets 13 – 14, 21, 28.

C2 Communication: C2 beacon, encrypted callback and C2 data exchange with 103.21.244.0. Packets 15 – 16, 25.

Escalation Details

Escalated To: Level II Incident Response Team

Escalation Reason: Multiple indicators of compromise were identified during Level 1 packet analysis including network reconnaisance/port scanning, possible SSH brute-force activity, malware download, suspicious outbound traffic, ransomware indicators, C2 communications, and DDoS traffic. Ransomware and C2 communications indicate a potentially active compromise and require further investigation by Level II Incident Response Team.

Escalation Time: Immediate

FINAL INCIDENT REPORT

WHAT HAPPENED?

During routine SOC monitoring, multiple suspicious network activities were identified in the provided packet capture. The evidence indicates a multi-stage network intusion involving reconnaissance through TCP SYN scanning and ICMP host discovery, followed by repeated SSH authentication failures. Additional activity included a suspicious executable download, suspicious outbound communications, possible C2 beaconing, possible data exfiltration, ransomware-related file encryption and DDoS attack.

HOW WAS THE ATTACK DETECTED?

The activity was detected through packet-level analysis in Wireshark. The investigation identified suspicious TCP SYN scans, repeated SSH login failures, HTTP requestes for suspicious executable, suspicious DNS queries, outbound connections, C2 beaconing, encrypted callback traffic, SMB file encryption activity, ransom note delivery, suspicious HTTP POST activity, and high-volume UDP broadcast traffic.

EVIDENCE SUPPORTING THE INCIDENT

Key evidence includes packets 4-6 showing repeated SSH failed login attempts, packets 7-8 showing malware.exe retrieval and file download, packets 11-12 showing .locked file encryption and ransomware note delivery, packets 15-16 showing C2 beaconing and encypted callback traffic, packet 10 showing suspicious data upload, packet 25 showing C2 data exchange, packets 13-14,21 and 28 showing DDoS related traffic.

RECOMMENDED ACTIONS

Isolate affected hosts, block malicious IP addresses through appropriate response process, reset potentially compromised credentials, initiate malware investigation/removal, monitor C2 communiocations, and apply approriate patches and security controls. The incident should be escalated to level II Incident Handling for further investigation and response

INTERPRETATION OF THE ATTACK

The packet evidences indicates a multi-stage network intrusion. The obeserved activity begins with reconnaissance and port scanning, followed by repeated SSH authentication failures. Suspicious executable downloads are observed, followed by C2 beaconing and data exchange. Ransomware related file encryption detected. Additional suspicious outbound traffic and DDoS activity are also present.

FINAL CONCLUSION

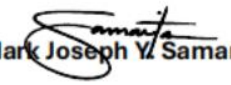
Multiple indicators of compromise were detected, indicating multi-stage network intrusion involving reconnaisance, brute force, malware activity, C2 communication, ransomware, suspicious outbound traffic, and DDoS activity. Due to the oberved ransomware and C2 indicators, the incident is classified as Critical and requires immediate escalation to the Level II Incident Response Team.

Result:

☐ Competent

☐ Not yet competent

Trainer:



Mark Joseph Y. Samarita